



ALFRED UNIVERSITY

SCHOOL OF ENGINEERING AND APPLIED TECHNOLOGY

Department of Applied Electronics and Embedded Technology

Embedded Systems + CETa Career Transition Program

Learning Resource Manual - Video-Enhanced Edition

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Learn. Build. Serve.

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Personal independent-study simulation. Alfred University is a fictional institution and this document is not an accredited university record.



Purpose

This guide is not a separate curriculum from your readiness checklist. The checklist is the syllabus; this document tells you where to learn each skill. Check a resource off only after you can use the knowledge in a real circuit, program, measurement, project, or troubleshooting exercise.

Primary rule: learn → build → measure → debug → document → then check the skill off.

VIDEO-ENHANCED EDITION: every learning domain now has a credible video or video-series path.

Written sources remain the reference for exact APIs, datasheets, and specifications.



How to Use This Guide

- ☐ Use the readiness checklist as the master list of what must be demonstrable. Do not try to “finish the internet” before building hardware.
- ☐ For most topics, use one primary resource, one visual/reference resource when needed, and one hands-on exercise. Avoid collecting multiple courses that teach the same thing.
- ☐ MUST resources support the common core or Project 1. SHOULD resources strengthen role-specific readiness. STRETCH resources should not delay the initial bridge-job application campaign unless a target posting specifically requires them.
- ☐ Keep personal bench practice on current-limited, safe low-voltage electronics. Follow tool/manufacturer safety instructions, especially for DMM current measurements and earth-referenced oscilloscope probes.

Recommended Learning Order

Stage	Focus
1	Electronics fundamentals + breadboard circuits
2	DMM + LTspice + basic measurements
3	C programming
4	STM32 GPIO / build / flash / debug
5	Bench supply + oscilloscope
6	Timers / interrupts / PWM
7	UART
8	Python ↔ UART automation
9	I2C OR SPI
10	Logic analyzer
11	Sensors + outputs / actuators
12	Git/GitHub + documentation
13	State machines + structured firmware
14	Project 1
15	Troubleshooting + test-report polish
16	Soldering / rework / harness proof



Stage	Focus
17	Complete ONE target-role add-on
18	Begin active applications
Later	CAN, FreeRTOS, second protocol, custom PCB, deeper Linux



Phase 1 - Electrical & Electronics Fundamentals

Build simple circuits as you learn. Calculation or video-only knowledge does not complete a hands-on checklist item.

Done	Resource	What to learn / how to use it	Priority
☐	All About Circuits - Free Textbook Free Primary textbook allaboutcircuits.com/textbook	Use as the main reference for DC circuits, Ohm's law, power, series/parallel networks, KVL/KCL, passive components, semiconductors, digital basics, and practical circuit labs. Read only the chapters that match your readiness checklist, then build and measure the circuit.	MUST
☐	Afretechmods - Electronics Tutorials Free Visual companion afrotechmods.com/tutorials	Use before the textbook when a concept feels abstract. Especially useful for voltage/current/resistance, dividers, LEDs, diodes, capacitors, transistors/MOSFETs, regulators, PWM, multimeters, and oscilloscopes.	MUST
☐	MIT OpenCourseWare 6.002 - Circuits and Electronics Free Deep reference ocw.mit.edu - 6.002	Do not complete this whole course before applying. Use individual lectures/notes when you want university-level depth on a checklist topic such as resistive networks, MOS devices, amplifiers, or dynamic circuits.	REF
☐	Analog Devices - LTspice Beginner Tutorial Free Simulation analog.com - LTspice tutorial	Learn schematic entry and waveform simulation. For each basic circuit: calculate it first, simulate it second, build it third, and compare measured results with theory.	MUST



Done	Resource	What to learn / how to use it	Priority
☐	All About Circuits - Digital Electronics Textbook Free Digital fundamentals allaboutcircuits.com/textbook/digital	Use for binary/hex, logic gates, digital voltage levels, pull-ups/pull-downs, CMOS/TTL concepts, timing, and introductory digital labs.	MUST
☐	Analog Devices - Op Amp Fundamentals Free Analog fundamentals analog.com - op amp fundamentals	Learn op-amp high-level behavior, buffering and amplification. Then breadboard a voltage follower and simple amplifier and observe input/output on the oscilloscope.	MUST

Phase 2 - C Programming & Embedded C

C fluency comes from writing and debugging code. Avoid tutorial copying; re-create examples from memory.

Done	Resource	What to learn / how to use it	Priority
☐	Beej's Guide to C Programming Free Primary C resource bej.us/guide/bgc	Work through the portions on types, functions, arrays, strings, pointers, structs, scope/storage duration, the preprocessor, bitwise operations, and memory. Write the examples yourself; do not copy/paste.	MUST
☐	Quantum Leaps - Modern Embedded Systems Programming Free Primary embedded-C course github.com/QuantumLeaps-modern-embedded-course	Use after basic C is comfortable. Learn ARM Cortex-M execution, memory-mapped I/O, registers, GPIO, interrupts, debugging, low-level C, and event/state-machine thinking. This bridges ordinary C into embedded C.	MUST
☐	Quantum Leaps - Video	Use as the video-course navigation companion to	REF



Done	Resource	What to learn / how to use it	Priority
	Course Hub Free Course navigation state-machine.com/video-course	the GitHub materials. Follow lessons that support the current checklist phase rather than trying to finish the entire advanced course first.	



Phase 3 - STM32 / ARM Cortex-M

Use official ST documentation early. The ability to find answers in datasheets/reference manuals is itself a required skill.

Done	Resource	What to learn / how to use it	Priority
☐	ST - STM32CubeIDE Basics MOOC Free Official ST course st.com - STM32CubeIDE MOOC	Learn project creation, GPIO, EXTI/interrupts, timers, PWM, ADC, DMA, USART, HAL/LL basics, flashing, and debugging. This is one of the main resources for Project 1.	MUST
☐	ST - STM32CubeMX & CubeHAL Basics MOOC Free Official ST course st.com - CubeMX & HAL MOOC	Learn pin configuration, clock-tree setup, peripheral configuration, code generation, HAL, callbacks, interrupts, DMA, GPIO, SPI, UART, timers, and ADC.	MUST
☐	ST - STM32 MCU Developer Resources Free Datasheets/reference manuals st.com - STM32 developer resources	Use this as the source for your board/MCU datasheet, reference manual, programming manual, application notes, errata, examples, and training. Practice finding answers in documentation instead of relying on tutorials.	MUST

Phase 4 - UART, I2C, SPI & CAN

UART is core. Project 1 needs at least one of I2C or SPI. CAN is role-driven/high-value rather than a universal first-application blocker.

Done	Resource	What to learn / how to use it	Priority
☐	SparkFun - Serial Communication / UART Free Protocol tutorial learn.sparkfun.com - serial communication	Learn asynchronous serial framing, baud rate, TX/RX wiring, voltage levels, and UART concepts. Then implement STM32 UART, use a terminal, connect Python, and decode the actual signal.	MUST



Done	Resource	What to learn / how to use it	Priority
☐	SparkFun - I2C Tutorial Free Protocol tutorial learn.sparkfun.com - I2C	Learn SDA/SCL, addresses, pull-ups, ACK/NACK, register transactions, and bus behavior. Hands-on is required if I2C is the serial bus chosen for Project 1 or your target role needs it.	MUST*
☐	SparkFun - SPI Tutorial Free Protocol tutorial learn.sparkfun.com - SPI	Learn SCLK, MOSI, MISO, chip-select, controller/peripheral roles, clock polarity/phase, bus speed, and transactions. Hands-on is required if SPI is the bus chosen for Project 1 or the target role needs it.	MUST*
☐	CSS Electronics - CAN Bus Simple Introduction Free Role-driven protocol csselectronics.com - CAN bus guide	Learn CAN_H/CAN_L, differential signalling, termination, identifiers, frames, arbitration, bit rate, error handling, and logging. Make this hands-on before applying to CAN-heavy automotive, robotics, industrial, energy, avionics, or HIL roles.	SHOULD



Phase 5 - Python for Hardware Test & Validation

This is a leverage area: use Python to command hardware, capture results, automate PASS/FAIL, and preserve evidence.

Done	Resource	What to learn / how to use it	Priority
☐	pySerial - Short Introduction Free Primary serial automation resource pyserial.readthedocs.io	Learn to open COM/tty ports, configure baud rate/timeouts, send bytes/commands, read responses, and cleanly close ports. Use it to build the Project 1 host/logger.	MUST
☐	pytest - Getting Started Free Test automation docs.pytest.org	Learn test functions, assertions, and simple fixtures/parameterization as needed. Translate hardware requirements into repeatable automated PASS/FAIL checks.	MUST
☐	Matplotlib - Quick Start Guide Free Test-data visualization matplotlib.org - quick start	Learn enough plotting to visualize sensor data, timing trends, current/voltage measurements, or failure patterns from your Python logs.	MUST

Phase 6 - Lab Equipment & Hands-On Electronics

Watch or read, then immediately practice on low-voltage hardware. Instrument confidence comes from use, not memorization.

Done	Resource	What to learn / how to use it	Priority
☐	Fluke - Digital Multimeter Basics Online Course Online course Primary DMM training fluke.com - DMM Basics	Learn voltage, resistance, continuity, current, diode-test functions, measurement symbols, specifications, accessories, and safe DMM habits. Follow it with hands-on measurements on low-voltage circuits.	MUST



Done	Resource	What to learn / how to use it	Priority
☐	Fluke - Digital Multimeter Essentials Optional paid course Certificate of completion fluke.com - DMM Essentials	Optional structured reinforcement if you want a more formal DMM course. Do not treat the completion certificate as a substitute for hands-on skill.	OPTION
☐	Tektronix - XYZs of Oscilloscopes Primer Free Primary scope reference tek.com - oscilloscope primer	Learn probes, volts/div, time/div, triggering, amplitude, period/frequency, duty cycle, coupling, noise/ripple, and general scope operation.	MUST
☐	EEVblog - Introduction to the Oscilloscope Free Visual scope companion eevblog.com - scope introduction	Use as the practical visual companion to the Tektronix primer. Practice every control on your own scope using safe low-voltage signals.	MUST
☐	Keysight - Bench Power Supply Basics Free Bench-supply fundamentals keysight.com - bench power supply basics	Learn voltage/current regulation, current limiting, constant-current behavior, readback, and deliberate first-power-up technique. Current-limited bring-up should become automatic.	MUST
☐	Saleae - Protocol Analyzers Free docs Logic-analyzer reference saleae.com/support/protocol-analyzers	Learn digital capture and protocol decoding for UART, I2C, SPI, and later CAN. Use actual captures to compare firmware intent with what the bus physically did.	MUST
☐	Adafruit - Guide to Excellent Soldering Free Primary soldering guide learn.adafruit.com - soldering	Learn tool setup, preparation, through-hole joints, headers/connectors, desoldering, common defects, and introductory SMD work. Practice on sacrificial boards/kits before valuable hardware.	MUST



Phase 7 - Wiring, Harnesses, Workmanship & ESD

Use professional workmanship standards as quality references. You do not need aerospace-operator certification for the bridge-job readiness threshold.

Done	Resource	What to learn / how to use it	Priority
☐	NASA-STD-8739.4 - Crimping, Interconnecting Cables, Harnesses, and Wiring Free standard Professional reference standards.nasa.gov - NASA-STD-8739.4	Use to see what high-quality cable/harness workmanship looks like: wire preparation, crimps, terminations, routing, strain relief, inspection, and documentation. Do not memorize aerospace criteria for Bridge Job #1.	REF
☐	NASA - Workmanship Standards Resources Free Workmanship/ESD reference sma.nasa.gov/workmanship	Use as a broader reference for professional electronics workmanship, soldering, cable harnesses, ESD controls, inspection, and reliability practices.	REF

Phase 8 - Git, GitHub, Linux & Development Workflow

Core Git is required; deeper Linux is role-driven. Your repos should show a clean engineering process and reproducible build/use instructions.

Done	Resource	What to learn / how to use it	Priority
☐	GitHub Skills - Introduction to Git Free Primary Git exercise github.com/skills/introduction-to-git	Learn commits, branches, history, and command-line Git basics in a guided exercise. Your core threshold is repository/status/diff/commit/push/.gitignore/README; deeper collaboration can grow later.	MUST
☐	GitHub Skills - Introduction to GitHub Free GitHub workflow github.com/skills/introduction-to-github	Learn repositories, branches, commits, pull requests, and the GitHub interface. Useful for building a clean public portfolio.	MUST



Done	Resource	What to learn / how to use it	Priority
☐	Pro Git - Free Book Free Reference book git-scm.com/book	Use as the authoritative reference when Git behavior is confusing. Do not read the entire book before starting your projects.	REF
☐	MIT - The Missing Semester (2026) Free Command line/dev tools missing.csail.mit.edu/2026	Use selected lessons for shell navigation, command-line tools, Git, debugging, packaging, and development environment habits. Linux depth is role-driven rather than a universal application blocker.	SHOULD



Phase 9 - Engineering Documentation, Test Discipline & Measurement Quality

Document what you built, expected, measured, failed, fixed, and verified. Evidence quality is a major interview differentiator.

Done	Resource	What to learn / how to use it	Priority
☐	GitHub Docs - About READMEs Free Portfolio documentation docs.github.com - READMEs	Use to structure project documentation so another person can understand what the project does, why it exists, how to build/run it, and where to find evidence/results.	MUST
☐	GitHub Docs - Issues Quickstart Free Engineering workflow docs.github.com - Issues	Use Issues or an equivalent structured project log to track bugs, tasks, design changes, and lessons learned. This makes your portfolio look like engineering work rather than a one-shot school demo.	SHOULD
☐	NIST - Measurement Uncertainty Basics Free Validation reference physics.nist.gov - uncertainty	Learn the practical meaning of accuracy, resolution, tolerance, repeatability, and uncertainty. Especially useful if you target hardware validation/test roles.	SHOULD

Phase 10 - Stretch / Role-Driven Growth

These skills are valuable but should not quietly move the application finish line. Complete them when your chosen role family or later embedded goals justify them.

Done	Resource	What to learn / how to use it	Priority
☐	FreeRTOS - Official Documentation Free RTOS reference docs.freertos.org	Only after bare-metal STM32 is comfortable. Learn tasks, priorities, queues, semaphores,	STRETCH



Done	Resource	What to learn / how to use it	Priority
		mutexes, ISR-to-task handoff, race conditions, and timing. Do not delay first-bridge applications solely for RTOS.	
☐	<u>KiCad - Getting Started in KiCad</u> Free PCB design <u>docs.kicad.org - getting started</u>	Learn schematic capture, ERC, footprints, PCB layout/routing, DRC, 3D viewing, BOM/fabrication outputs, and board-release workflow for the optional custom-PCB differentiator project.	STRETCH
☐	<u>Phil's Lab - STM32 PCB Design Full Tutorial</u> Free video PCB design companion <u>youtube.com - Phil's Lab STM32 PCB tutorial</u>	Use after basic KiCad. Follow the full MCU-board workflow: power, decoupling, clocks, reset/boot, SWD, connectors, layout, routing, ERC/DRC, BOM, and manufacturing outputs.	STRETCH

Project 1 - Resource Bundle

Use Project 1 as the practical exam for the common core. You are not finished because you watched the resources; you are finished when the project demonstrates the skill and you can explain the evidence.

Project 1 area	Use these resources
Electronics & sensor interfacing	All About Circuits + Afrotechmods + datasheet
C & firmware structure	Beej + Quantum Leaps
STM32 setup/peripherals	ST CubeIDE MOOC + CubeMX/HAL MOOC
UART/I2C/SPI	SparkFun + Saleae captures
Python host/logger	pySerial + pytest + Matplotlib
DMM/scope/bench supply	Fluke + Tektronix/EEVblog + Keysight
Version control	GitHub Skills + Pro Git



Project 1 area	Use these resources
Documentation	GitHub README guidance + your readiness checklist
Troubleshooting proof	Your own debug logs, measurements, captures, and regression tests

Weekly Study Rhythm

A sustainable template while working full-time:

- ☐ Session 1 (60-90 min): learn one concept from the primary resource.
- ☐ Session 2 (60-90 min): code, simulate, or solve examples without copying.
- ☐ Session 3 (45-60 min): review, explain the concept aloud, and prepare the lab.
- ☐ Weekend block (2-3 hours): physically build, measure, debug, and document evidence.

Application Trigger Reminder

START ACTIVE APPLICATIONS when: the common core is demonstrable, Project 1 is interview-ready, and ONE role-specific add-on is substantially complete. Do not wait for every SHOULD/STRETCH resource, the complete embedded-engineer skill set, or the associate degree solely because waiting feels safer.

Quick Link Index

All resource names below are clickable. This index is for fast digital access; the earlier sections explain how each resource fits the plan.

1: Electrical & Electronics Fundamentals

- [All About Circuits - Free Textbook](#)
- [Afrotechmods - Electronics Tutorials](#)
- [MIT OpenCourseWare 6.002 - Circuits and Electronics](#)
- [Analog Devices - LTspice Beginner Tutorial](#)
- [All About Circuits - Digital Electronics Textbook](#)
- [Analog Devices - Op Amp Fundamentals](#)

2: C Programming & Embedded C

- [Beej's Guide to C Programming](#)
- [Quantum Leaps - Modern Embedded Systems Programming](#)
- [Quantum Leaps - Video Course Hub](#)

3: STM32 / ARM Cortex-M

- [ST - STM32CubeIDE Basics MOOC](#)
- [ST - STM32CubeMX & CubeHAL Basics MOOC](#)
- [ST - STM32 MCU Developer Resources](#)



4: UART, I2C, SPI & CAN

- [SparkFun - Serial Communication / UART](#)
- [SparkFun - I2C Tutorial](#)
- [SparkFun - SPI Tutorial](#)
- [CSS Electronics - CAN Bus Simple Introduction](#)

5: Python for Hardware Test & Validation

- [pySerial - Short Introduction](#)
- [pytest - Getting Started](#)
- [Matplotlib - Quick Start Guide](#)

6: Lab Equipment & Hands-On Electronics

- [Fluke - Digital Multimeter Basics Online Course](#)
- [Fluke - Digital Multimeter Essentials](#)
- [Tektronix - XYZs of Oscilloscopes Primer](#)
- [EEVblog - Introduction to the Oscilloscope](#)
- [Keysight - Bench Power Supply Basics](#)
- [Saleae - Protocol Analyzers](#)
- [Adafruit - Guide to Excellent Soldering](#)

7: Wiring, Harnesses, Workmanship & ESD

- [NASA-STD-8739.4 - Crimping, Interconnecting Cables, Harnesses, and Wiring](#)
- [NASA - Workmanship Standards Resources](#)

8: Git, GitHub, Linux & Development Workflow

- [GitHub Skills - Introduction to Git](#)
- [GitHub Skills - Introduction to GitHub](#)
- [Pro Git - Free Book](#)
- [MIT - The Missing Semester \(2026\)](#)

9: Engineering Documentation, Test Discipline & Measurement Quality

- [GitHub Docs - About READMEs](#)
- [GitHub Docs - Issues Quickstart](#)
- [NIST - Measurement Uncertainty Basics](#)

10: Stretch / Role-Driven Growth

- [FreeRTOS - Official Documentation](#)
- [KiCad - Getting Started in KiCad](#)
- [Phil's Lab - STM32 PCB Design Full Tutorial](#)

COMPLETE VIDEO COMPANION

Use this section alongside Phases 1-10 above. A video is not a substitute for the hands-on proof standard: watch the indicated material, then build, measure, code, debug, and explain the concept without copying a finished solution.



Coverage rule: where one short video cannot responsibly cover the bridge-job threshold, the link is a series/hub or the row contains multiple linked videos. Watch the modules named in the coverage column. MUST topics support the common core; SHOULD/ROLE/STRETCH items do not move the application finish line unless a target posting makes them essential.

Video Phase 1 - Electrical & Electronics Fundamentals

Complete these visual lessons while physically reproducing the basic circuits. The All About Circuits hubs are intentionally broad series, not single clips.

Done	Topic / priority	Video or video series	Bridge-job coverage standard
☐	MUST - DC quantities, Ohm/Kirchhoff/power, series/parallel, dividers, opens/shorts	All About Circuits - Video Lectures	Watch Basic Electronics, Circuits, Ground/Reference, and Circuit Troubleshooting sections. Be able to calculate and then verify simple circuits with a DMM.
☐	MUST - Passive components, RC behavior, filters, ratings/tolerance	All About Circuits - Video Tutorials	Use the components/circuit-elements lessons for resistors, capacitors, inductors, RC behavior and basic filtering. Pair each lesson with a breadboard or LTspice exercise.
☐	MUST - Diodes, LEDs, BJTs, MOSFETs, switching/protection	All About Circuits - Video Lectures	Watch Diodes and Transistor Circuits sections. Focus on polarity, forward voltage, current limiting, transistor-as-switch behavior and MOSFET low-side switching.
☐	MUST - Digital logic, binary/hex, gates and logic levels	All About Circuits - Video Lectures	Watch the Digital section: digital concepts, binary conversion, logic gates and truth tables. Hex/bit-mask fluency is reinforced again in embedded C.
☐	SHOULD - Op-amp fundamentals and simple signal conditioning	TI Precision Labs - Op Amps	For the bridge-job threshold, prioritize introduction, input/output operating limits, buffering/gain, bandwidth/slew concepts and common practical limitations; advanced precision design can wait.
☐	SHOULD - Comparators and hysteresis	TI Precision Labs - Comparators	Watch comparator functions, noise/hysteresis and op-amp-vs-comparator lessons. You should be able to explain thresholding and why hysteresis prevents chatter.
☐	MUST - Circuit simulation with LTspice	Analog Devices - LTspice Basics Video Series	Complete installation/schematic capture, transient simulation and AC-analysis basics. Use simulation to predict a circuit before measuring the real build.
☐	MUST - Reading schematics and tracing signals/power	DigiKey - How to Read a Schematic	Learn symbols, nets, power/ground, connectors and signal tracing. Then trace your own Project 1 power tree and one MCU signal from pin to peripheral.
☐	MUST - Reading datasheets, limits, pinouts and application circuits	DigiKey - How to Read a Datasheet	Learn to locate pinouts, maximum ratings, recommended operating conditions, electrical characteristics and example circuits. Practice on the exact MCU and sensor datasheets you use.



Video Phase 2 - C Programming & Embedded C

C is a demonstrable skill, not a watch-time requirement. Re-type small examples, then close the tutorial and recreate them from memory.

Done	Topic / priority	Video or video series	Bridge-job coverage standard
☐	MUST - C language foundations: types, functions, arrays, strings, structs and pointers	freeCodeCamp - C Programming Tutorial for Beginners	Use the full course for core syntax and memory/pointer foundations. Skip basic sections only if you can already code the exercises from memory.
☐	MUST - Embedded C: fixed-width data, bitwise masks, volatile, registers and memory-mapped I/O	Quantum Leaps - Modern Embedded Systems Programming	Follow the C/ARM lessons that repeatedly move between C and the MCU. This is the main video path for bit operations, registers, memory, hardware access and low-level reasoning.
☐	MUST - Interrupt-aware, non-blocking code, state machines and debugging	Quantum Leaps - Modern Embedded Systems Programming	Use the interrupt/event/state-machine/debug lessons. Build at least one non-blocking state machine and explain why long delay loops are poor architecture for responsive firmware.

Video Phase 3 - STM32 / ARM Cortex-M

Use official ST video training as the primary path, then use the MCU/Quantum Leaps series to understand what the configuration tools are doing underneath.

Done	Topic / priority	Video or video series	Bridge-job coverage standard
☐	MUST - STM32CubeIDE workflow, GPIO, EXTI, timers/PWM, ADC/DMA and USART	STMicroelectronics - STM32CubeIDE Basics MOOC	Complete the hands-on modules relevant to GPIO, interrupts, timer/PWM, ADC/DMA and USART. Build, flash and debug each exercise instead of only watching.
☐	MUST - STM32CubeMX, pin/clock configuration, HAL, callbacks and peripherals	STMicroelectronics - STM32CubeMX & CubeHAL Basics MOOC	Watch the pinout, clock tree, peripheral configuration, HAL, callbacks, interrupts and DMA sections plus GPIO/SPI/UART/timer/ADC exercises.
☐	MUST - MCU architecture: CPU/core, clocks, memory, timers, watchdog, ADC and debug mindset	TI Precision Labs - Microcontrollers Video Series	Watch What is an MCU, clocking, memory, timer/PWM, watchdog, ADC and Debug Mindset modules. These concepts transfer directly to STM32 even though the examples may use TI MCUs.
☐	MUST - SWD/debugger reasoning, registers and fault investigation	Quantum Leaps - Modern Embedded Systems Programming ST - STM32CubeIDE Basics MOOC	Use breakpoints, stepping, variables/register views and fault reasoning in your own project. The goal is to diagnose instead of repeatedly reflashing or power-cycling.
☐	MUST - Sensors/actuators with ADC, PWM, timers and safe GPIO interfacing	TI Precision Labs - Microcontrollers Video Series ST - STM32CubeIDE Basics MOOC	Use the ADC/PWM/timer lessons, then implement a real sensor input and controlled output. Respect GPIO voltage/current limits and use a driver transistor where the load requires it.



Video Phase 4 - UART, I2C, SPI, CAN & Role-Driven Interfaces

UART is universal core. For Project 1, become hands-on with I2C OR SPI. The second bus and CAN/RS-485/Ethernet can continue while you are already applying unless your role track requires them.

Done	Topic / priority	Video or video series	Bridge-job coverage standard
☐	MUST - UART framing, baud rate, wiring, error sources and MCU operation	TI Precision Labs - Microcontroller Communication	Watch both UART modules: MCU overview plus protocol/error overview. Then implement terminal communication and Python-to-MCU commands and decode a frame on your logic analyzer.
☐	MUST - I2C: open-drain, pull-ups, addressing, ACK/NACK, timing and transactions	TI Precision Labs - I2C Video Series	If I2C is your Project 1 bus, watch the hardware/protocol modules and the advanced pull-up/timing material needed to troubleshoot NACKs or a stuck bus.
☐	MUST - SPI: lines, master/peripheral roles, modes, timing and STM32 use	TI Precision Labs - Microcontroller Communication DigiKey - STM32/Nucleo SPI Hands-On	Watch SPI overview, modes and timing, then the STM32 implementation. Capture one transaction and correctly identify SCLK, MOSI, MISO, CS and the selected mode.
☐	SHOULD - CAN/CAN-FD physical layer, framing, termination and logging concepts	TI Precision Labs - CAN/LIN/SBC Series CSS Electronics - CAN Bus Intro + Video	Use TI for physical/protocol fundamentals and CSS for practical logging/decoding context. Make this hands-on before applying to CAN-heavy automotive, robotics, industrial, energy, avionics or HIL roles.
☐	ROLE - RS-485 differential signaling, termination and multi-drop basics	TI Precision Labs - RS-485 Series	Role-driven. Watch all three short modules if a posting mentions RS-485/Modbus/industrial serial. Know why a transceiver and termination/biasing are needed.
☐	ROLE - Ethernet PHY/link basics for hardware troubleshooting	TI Precision Labs - Ethernet Series	Role-driven. For the bridge threshold, focus on what an Ethernet PHY is, clocks and a basic 100BASE-TX schematic; deeper high-speed SI design is not required.

Video Phase 5 - Python for Hardware Test & Validation

This section assumes normal Python programming fluency and focuses on turning Python into a hardware-test tool: serial I/O, evidence, repeatable PASS/FAIL and clean reporting.

Done	Topic / priority	Video or video series	Bridge-job coverage standard
☐	MUST - pySerial: COM/tty setup, baud/frame settings, send/receive and timeouts	Python + PySerial Microcontroller Tutorial	Follow the serial-port sections covering installation, port selection, configuration, transmit, receive, timeout and bidirectional communication. Repeat it with your STM32 rather than Arduino.
☐	MUST - Python-driven instrument/device automation mindset	Keysight - Control and Automate Instruments in Python	Use this to learn the professional automation pattern: configure instrument/device, send commands, acquire measurements, structure scripts and automate repeatable test workflows.
☐	MUST - Structured logging and useful diagnostic records	Corey Schafer - Python Logging Basics	Learn levels, file logging and formatting. Apply it to timestamp test name, expected/actual, PASS/FAIL, exceptions and DUT/firmware version.



Done	Topic / priority	Video or video series	Bridge-job coverage standard
☐	MUST - CSV and JSON test-result storage	Corey Schafer - CSV Module Corey Schafer - JSON Module	Be able to write/read both formats and choose one appropriate for measurement tables vs structured nested results. Use at least one in Project 1.
☐	SHOULD - Command-line arguments and externalized configuration	Real Python - argparse Video Course	Build a small CLI that accepts port/device/test options. Keep thresholds/addresses/settings outside hard-coded test logic when practical.
☐	MUST - Automated tests, assertions and repeatable PASS/FAIL	Harvard CS50P - pytest Short	Learn test functions, assertions and basic test organization. Use these ideas to test parsers/calculations and to keep DUT failures separate from script bugs.
☐	MUST - Measurement plots and time-series visualization	Matplotlib - Anatomy of Matplotlib	Learn figure/axes/plot basics, labels and saving output. Create at least one useful sensor/time-series or test-summary plot, not decorative graphics.

Video Phase 6 - Lab Equipment & Hands-On Electronics

Immediately practice every instrument lesson on safe, current-limited low-voltage circuits. Do not check these off based on watching alone.

Done	Topic / priority	Video or video series	Bridge-job coverage standard
☐	MUST - Digital multimeter: voltage, resistance, continuity, current and diode test	Fluke - Digital Multimeter Basics Course	Complete the DMM lessons and practice correct lead/jack/mode selection. You must understand series current measurement and never place a current-configured meter across a source.
☐	MUST - Bench power supply: voltage/current limits, CV/CC and bring-up	Keysight - Bench Power Supply Basics	Prioritize voltage/current/regulation, constant-voltage vs constant-current mode, current limiting and safe first power-up. Measure current in at least two Project 1 operating states.
☐	MUST - Oscilloscope controls, probes, trigger, amplitude and timing	EEVblog - Introduction to the Oscilloscope	Practice x10 probe use, volts/div, time/div, edge trigger, frequency/period, amplitude/duty cycle and capture of PWM/clock/serial-related signals.
☐	MUST - Oscilloscope earth-ground safety	EEVblog - How NOT to Blow Up Your Oscilloscope	Understand that a typical bench-scope reference clip is earth-referenced. Keep personal work low-voltage and know when differential/isolated methods are required.
☐	MUST - Logic analyzer capture and UART/I2C/SPI protocol decoding	Saleae - Using Protocol Analyzers	Watch the embedded tutorial, then capture/interpret your own UART and chosen I2C/SPI traffic. Configure sample rate/decoder correctly and export evidence.
☐	MUST - Soldering, desoldering/rework and inspection	PACE - Basic Soldering Lesson 1 (continue series) Adafruit - Collin's Lab: Soldering	Use PACE as the workmanship series and Adafruit as the visual practical companion. Practice through-hole, headers, wire prep, desoldering and basic larger SMD; inspect for bridges/cold joints.



Video Phase 7 - Wiring, Harnesses, Workmanship & ESD

These videos build professional bench habits. They are not substitutes for employer-specific workmanship training or formal certification.

Done	Topic / priority	Video or video series	Bridge-job coverage standard
☐	MUST - Wire stripping, crimping and quality terminations	TE Connectivity - A Primer on Small Wire Crimping	Watch the 60-minute webinar, then make and continuity-test several crimped prototype leads. Be able to explain conductor/terminal/tool compatibility and basic quality checks.
☐	MUST - Harness/workmanship mindset and controlled electronics handling	NASA - Workmanship Program (videos + standards)	Use NASA's workmanship page as the visual/quality reference. Focus on orderly assembly, strain relief, inspection discipline, personnel grounding and why workmanship affects reliability.
☐	MUST - ESD causes, damage mechanisms and prevention	NASA - Electrostatic Discharge Overview (on Workmanship page) EOS/ESD Association - Introductory ESD Lecture	Understand invisible ESD damage, personal/workstation grounding and basic EPA behavior. Then set up and use your own ESD-safe work practice for MCU/PCB handling.

Video Phase 8 - Git, GitHub, Linux & Development Workflow

The bridge-job target is a reproducible project workflow, not DevOps mastery.

Done	Topic / priority	Video or video series	Bridge-job coverage standard
☐	MUST - Git/GitHub repositories, commits, branches and pull-request workflow	GitHub - Git and GitHub Learning Resources / Beginners Series GitHub Skills - Introduction to GitHub	Use the official beginner video series plus the interactive exercise. Core application threshold: clone/init, status/diff, meaningful commits, push, .gitignore and readable README.
☐	MUST - Shell navigation, files, pipes/streams, environment and CLI conventions	MIT - Missing Semester 2026 Video Lectures	Watch Introduction to the Shell and Command-line Environment. Practice pwd/ls/cd/mkdir/cp/mv/rm, cat/less/head/tail/grep and safe command-line habits.
☐	SHOULD - Development tools, debugging/profiling, Git depth and code quality	MIT - Missing Semester 2026 Video Lectures	Watch Development Environment and Tools, Debugging and Profiling, Version Control and Git, Packaging/Shipping and Code Quality as needed. Do not delay applications for advanced shell mastery.

Video Phase 9 - Troubleshooting, Documentation, Test Discipline & Measurement Quality

This phase turns "I built it" into professional evidence: reproduce, measure, isolate, document, verify and communicate.

Done	Topic / priority	Video or video series	Bridge-job coverage standard
☐	MUST - Systematic electronics troubleshooting and root-cause isolation	All About Circuits - Troubleshooting Video Lectures EEVblog - Circuit Analysis & Debugging Example	Watch system-level and circuit troubleshooting sections. Practice power-first checks, block isolation, signal tracing, substitution and one-change-at-a-time diagnosis; document three real failure stories.



Done	Topic / priority	Video or video series	Bridge-job coverage standard
☐	MUST - Project documentation: photos, video, written build/process evidence	DigiKey - Documenting Your Electronics Projects	Use the video for capture/documentation habits. Your engineering package must still include block diagram, schematic/wiring, requirements, build instructions, tests, results, known issues and Rev-B ideas.
☐	MUST - Requirement -> test -> expected/actual -> PASS/FAIL -> regression	Speedgoat - Automated and Continuous HIL Testing Webinar	Use the HIL webinar for the professional testing concepts: why controlled tests exist, repeatability, automation, regression and continuous testing. Ignore vendor-specific HIL depth unless you select the HIL role track.
☐	SHOULD - Accuracy, precision, resolution and repeatability	Fluke - Why DMM Accuracy and Precision Matter	Be able to explain accuracy vs precision vs resolution and why repeatable readings are not necessarily accurate. Apply this when interpreting bench measurements.
☐	SHOULD - Measurement uncertainty and risk concept	Keysight - Measurement Uncertainty and Risk	Bridge-level goal: understand that reported measurements have uncertainty and that uncertainty affects pass/fail risk. Detailed metrology calculations are not a first-bridge blocker.

Video Phase 10 - Stretch / Role-Driven Growth

These links make the guide complete without turning optional growth into prerequisites. Only promote one to MUST when your target role explicitly depends on it.

Done	Topic / priority	Video or video series	Bridge-job coverage standard
☐	STRETCH - FreeRTOS: tasks, scheduling, queues, mutexes, semaphores and interrupts	Shawn Hymel / DigiKey - Introduction to RTOS Video Course	Follow Parts 1-11 as needed; complete the challenges. Learn RTOS after bare-metal STM32/state-machine fundamentals are comfortable.
☐	STRETCH - Custom STM32 PCB in KiCad	Phil's Lab - STM32 PCB Design Full Tutorial	Watch the full schematic-to-Gerber workflow: decoupling, clocks, power, SWD, layout, routing, ERC/DRC and manufacturing outputs. Build your own design rather than publishing an exact copy.
☐	ROLE - Hardware-in-the-loop / automated validation architecture	Speedgoat - Automated and Continuous HIL Testing Webinar	For Hardware Validation/Embedded Test/HIL targeting: understand stimulus -> DUT -> measurement -> pass/fail, fault injection, regression and CI concepts. Build Project 2 rather than buying a commercial HIL rack.
☐	SHOULD - Basic C++ familiarity for firmware-heavy postings	freeCodeCamp - C++ Tutorial for Beginners	Only learn enough to read and modify straightforward C++ firmware at this stage: classes/objects, constructors, references/pointers and basic inheritance. Do not let this replace C/STM32 mastery.
☐	ROLE - RS-485 and Ethernet depth when postings request them	TI Precision Labs - RS-485 Series TI Precision Labs - Ethernet Series	Treat as role-driven. If selected, demonstrate at least basic troubleshooting vocabulary and physical-interface understanding rather than merely listing the protocols on a resume.



Project 1 - Video Stack and Build Rule

Project 1 is intentionally NOT linked to a single finished “STM32 weather station” tutorial. Copying one complete project would weaken the portfolio evidence. Use the component-level videos below, then integrate the system yourself.

Done	Topic / priority	Video or video series	Bridge-job coverage standard
☐	MUST - Circuit/sensor/power foundation	All About Circuits - Video Lectures DigiKey - Read a Datasheet	Design and verify the sensor/power/output circuits instead of copying a wiring diagram blindly.
☐	MUST - STM32 firmware and peripherals	ST - STM32CubeIDE Basics MOOC ST - CubeMX/CubeHAL Basics MOOC	Build GPIO + timer/interrupt + UART + your chosen I2C/SPI bus with modular C and explicit error/timeout behavior.
☐	MUST - UART + I2C or SPI bus understanding	TI - MCU Communication Series TI - I2C Series	Understand the protocol before using HAL calls. Decode the real bus on a logic analyzer.
☐	MUST - Python host/logger/test tool	PySerial Microcontroller Tutorial Harvard CS50P - pytest Matplotlib Video	Command the DUT, parse responses, log data, run explicit PASS/FAIL checks and generate a useful plot/summary.
☐	MUST - Lab measurements and evidence	Fluke - DMM Basics EEVblog - Oscilloscope Intro Saleae - Protocol Analyzers	Measure rails/current, capture a waveform, decode UART + your chosen serial bus and save labeled evidence.
☐	MUST - Debug + document the real failures	All About Circuits - Troubleshooting Videos DigiKey - Documenting Projects	Produce at least one physical/electrical fault story plus firmware/interface faults with symptom -> evidence -> root cause -> fix -> regression proof.

Video Coverage Audit

- ☐ Phase 1 electronics, circuit reading, datasheets, simulation and troubleshooting have video paths.
- ☐ Phase 2 C and embedded-C concepts have video paths.
- ☐ Phase 3 STM32/MCU architecture, peripherals and debugging have video paths.
- ☐ Phase 4 UART/I2C/SPI/CAN plus role-driven RS-485/Ethernet have video paths.
- ☐ Phase 5 Python serial I/O, automation, logging, file output, CLI, pytest and plotting have video paths.
- ☐ Phase 6 DMM, power supply, oscilloscope, scope-ground safety, logic analyzer and soldering have video paths.
- ☐ Phase 7 crimping/harness workmanship and ESD have video paths.
- ☐ Phase 8 Git/GitHub, shell/Linux and development-tool workflow have video paths.
- ☐ Phase 9 troubleshooting, documentation, test discipline and measurement-quality concepts have video paths.
- ☐ Phase 10 FreeRTOS, custom PCB, HIL, C++ and role-driven interface depth have video paths.
- ☐ Project 1 has a complete component-level video stack without encouraging a copied end-to-end project.

Final use rule: do not wait until every SHOULD/ROLE/STRETCH video is complete. Start active applications when the readiness checklist common core, Project 1, and ONE target-role add-on are demonstrable. Continue the deeper video paths while interviewing.